CHIMATA NAGA GOPAL

Python Developer | AI / ML Engineer

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SUMMARY

Energized by hard problems and curious by nature, I love turning a plain-language idea into something people can see and use. I build AI agents and generative pipelines that take a simple brief and return finished posters, images, and videos — so people spend less time on repetitive production and more on their ideas. With deep learning research roots and full-stack skills to ship, I move from experiment to product with ease, adapt to any team's rhythm, and care as much about people as I do about the technology.

SKILLS

GEN AI / LLMS	AI Agents / RAG / NLP / Transformers
ML / DL	CNN / LSTM / ANN / RNN / Vision Transformer (ViT) / Attention Mechanisms / LoRA Fine-Tuning
GENERATIVE	Diffusion-Based Image Generation / Open-Source Video Generation / Background Removal / Image Upscaling
CV & MEDIA	OpenCV / YOLO / MediaPipe / FFmpeg / Image Processing / NumPy / Pandas / Matplotlib
BACKEND	Python / Node.js / Express.js / REST APIs / FastAPI / Async Job Processing / SQL / MySQL
FRONTEND & TOOLS	React.js (Basics) / JavaScript / HTML / CSS / Bootstrap / Git / GitHub / Postman / Jupyter Notebook

EXPERIENCE

Daksh Global Innovations Pvt Ltd *(formerly TechPixe)* — **NOV 2025 – PRESENT**

Junior AI & ML Developer | Hyderabad, India

AI Creative Studio Agent — end-to-end Generative AI platform

- Built an AI-agent platform that turns natural-language briefs into product images, promotional posters, and short marketing videos — one request yields a full campaign asset set.
- Designed the agent's plan-and-execute workflow: understand the request, plan the steps, select generation tools, run them, and validate outputs before delivery.
- Implemented business-specific RAG over products, offers, pricing, and brand guidelines so generated content reflects current campaign data and brand rules.
- Fine-tuned open-source image-generation models with LoRA to reproduce specific products, brands, and visual styles for each business.
- Integrated open-source image and video generation models with FFmpeg processing, background removal, upscaling, and format conversion.
- Ran generation as async GPU jobs with automated quality validation; built the web UI to manage and download assets.

FO2.AI – AI-Powered Content & Image Generation Platform

- Built an AI-powered platform where users enter natural-language prompts to generate context-aware content and images using OpenAI APIs.
- Designed prompt-engineering logic that interprets user inputs and generates optimized prompts for accurate content and image output.
- Implemented subscription-based access with Free, Trial, and Premium plans including per-tier AI usage tracking.
- Built backend APIs for user authentication, subscription management, and AI request handling.

PROJECTS

CNN-Trans: Parallel Dual-Branch Network for Fundus Image Classification *(Academic / Research)*

- Designed a parallel dual-branch network: CNN-LSTM (local features) + Vision Transformer (global features) for fundus classification.
- Added Coordinate Attention (CNN-LSTM) and self-attention (ViT) to sharpen features and suppress redundancy.
- Achieved 80.68% accuracy, competitive with state-of-the-art methods.
- Targeted early detection of retinal diseases such as diabetic retinopathy and glaucoma, enabling faster clinical diagnosis through automated fundus image analysis.

EDUCATION & CERTIFICATIONS

Usha Rama College of Engg. & Technology
B.Tech, ECE | CGPA 8.59
2021 – 2025

N R I Junior College, Guntur
Intermediate | 88.8%
2019 – 2021

AI/ML Virtual Internship
Virtual Internship Certificate

Python Course
Udemy | Certificate of Completion